

For notational simplicity, we set

$$\cdot \hbar = c = 1$$

In special relativity, our universe is represented by the vector space $\mathbb{R}^{1,3}$

$$\cdot \mathbb{R}^{1,3} := \mathbb{R}^4 \text{ with a different inner product.}$$

$$\dots \text{ or } \mathbb{R}^{3,1}$$

Lecture 9 (Special relativity)

• $\hbar = c = 1$ • Our universe = $\mathbb{R}^{1,3} = \mathbb{R}^4$ with a different inner product

• $x \in \mathbb{R}^{1,3}$ • $x = (x^0, x^1, x^2, x^3)$, where x^0 is the time coordinate and $x := (x^1, x^2, x^3)$ is the 3-tuple of spatial coordinates

• We define a symmetric matrix η as
$$\eta = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

• η_{uv} is the entry of η

• Given $x, y \in \mathbb{R}^{1,3}$, $(x, y) = x^0 y^0 - x^1 y^1 - x^2 y^2 - x^3 y^3 = \sum_{u,v} \eta_{uv} x^u y^v$ (Minkowski 423)

• $x \cdot y = x^1 y^1 + x^2 y^2 + x^3 y^3$, thus $(x, y) = x^0 y^0 - x \cdot y$ (standard 423)

• In special relativity, $x^2 = (x, x) = (x^0)^2 - x \cdot x$ • Euclidean norm of $x := |x|$

$$(x^0)^2 - x \cdot x = x^2$$

On the other hand, L (Lorentz transformation) on $\mathbb{R}^{1,3}$ is a linear map s.t. $(Lx, Ly) = (x, y)$

$\forall x, y \in \mathbb{R}^{1,3}$ • $\Rightarrow L^T \eta L = \eta$ (this is usual inner product orthogonal transformation)

The set of Lorentz transformations = $O(1,3)$

We have the following two homomorphisms from L .

$$(i) \det : O(1,3) \rightarrow \mathbb{Z}_2 = \{1, -1\}$$

$$(ii) L \mapsto \text{sign}(L_0^0), \text{ where } L_0^0 \text{ is the top left entry of the matrix}$$

$$SO^+(1,3) := \{ L \in O(1,3) \mid \det(L) = 1, \text{ and } \text{sign}(L_0^0) = 1 \} \subset O(1,3)$$

□

The parametrization of a curve in Euclidean space by arc length has the property that it is invariant under Euclidean symmetries. That is, under rotation and translation, to the coordinate system, the parametrization remains unchanged.

~~As in Euclidean space~~, In a similar way to Euclidean space,

a curve in $\mathbb{R}^{1,1}$ can sometimes be parametrized in a way that is invariant under any change of the coordinate system by an action of $SO(1,1)$ or a spacetime translation.

Euclidean space vs Minkowski space

In a similar vein

Euclidean norm

$$\left(\left| \frac{dx}{d\tau} \right|^2 = 1 \right) \rightarrow \left(\left| \frac{dx}{d\tau} \right|^2 - \left| \frac{dt}{d\tau} \right|^2 = -1 \right)$$

A sufficient condition is that in some coordinate system, the curve is a time parameterization $X(t) = (t, X(t))$ satisfying $\left| \frac{dX}{dt} \right| < 1$. If

that is, in some coordinate system, the curve represents the trajectory of a particle whose speed is always strictly less than c .

Def: If $X(\tau)$ satisfies $\left| \frac{dX}{d\tau} \right| = 1$, we call τ as 'proper time'.

$X(\tau)$, τ : proper time. Mass of particle = m then 'four-momentum' at any point on the trajectory is $p = m \frac{dX}{d\tau}$

Note, p does not depend on the coordinate system

In a fixed coordinate system, the 'four-momentum' can be written

$$p = \left(\frac{m}{\sqrt{1-v^2}}, \frac{mv}{\sqrt{1-v^2}} \right), \quad v(t) := \text{coordinate-dependent velocity}$$

$$p = (p^0, p^1, p^2, p^3) = (E, \mathbf{p}) \quad (\text{relativistic momentum})$$

$$(E, \text{relativistic energy})$$

$$p^2 = (p^0)^2 - |\mathbf{p}|^2 = m^2 \quad \therefore p^0 = E = \sqrt{m^2 + p^2} = W_p$$

We suppress the dependence on the relativistic measure

~~The~~ The fourth momentum is always an element of the 4D world

$$X_m = \{p \in \mathbb{R}^{1,3} : p^2 = m^2, p^0 \geq 0\}$$

This manifold is called the mass shell of mass m .

~~Lecture 10 (The mass shell)~~

Lecture 10 (The mass shell)

$H = L^2(X_m, d\lambda_m)$, where λ_m is the unique measure that is invariant under the action of the restricted Lorez G-P on X_m

How to define λ_m ? [Lub, Miguel, Introduction] Lectures on Quantum field theory

Given an invariant function $f(p)$ of four momentum p of a particle mass m with $p^0 > 0$, $\exists$ integration measure

$$\int \frac{d^4 p}{(2\pi)^4} \delta(p^2 - m^2) \Theta(p^0) f(p)$$

where Θ is the Heaviside step fcn.

manifestly Lorez invariant for Lorez transformation

By using $\delta[f(x)] = \sum_{x_i = \text{zeros of } f} \frac{1}{|f'(x_i)|} \delta(x - x_i)$, we have

$$\delta(p^2 - m^2) = \frac{1}{2p^0} \delta(p^0 - \sqrt{|\mathbf{p}|^2 + m^2}) + \frac{1}{2p^0} \delta(p^0 + \sqrt{|\mathbf{p}|^2 + m^2})$$

$$\textcircled{AB} = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\sqrt{|\mathbf{p}|^2 + m^2}} f(\sqrt{|\mathbf{p}|^2 + m^2}, \mathbf{p})$$

So the relativistic invariant measure is given by

$$\int \frac{d^3 p}{(2\pi)^3} \frac{1}{W_p} \quad \text{with } W_p = \sqrt{|\mathbf{p}|^2 + m^2}$$

$$\text{or use } \delta[(x-a)(x-b)] = \frac{1}{|b-a|} [\delta(x-a) + \delta(x-b)]$$

Lecture 11 (The postulates of quantum field theory)

Q What does it mean to say momentum state at a given time?

How to change the postulates in special relativity setting?

Recall that we had five postulates of quantum mechanics that were introduced in Lec 2.

~~P1~~ Lorentz transforms changes the concepts of time slices and the notion of two events happening 'at the same time' need not remain the same under a change of coordinate system.

The postulates need to be modified as follows.

~~P1~~ For any quantum system, $\exists$ H (Hilbert space) s.t. the state of the system is described by vectors in H . A state is not for a given time but for all space time. A state gives a complete spacetime description of the system.

~~P2~~ unchanged

~~P3~~ unchanged

~~P4~~ unchanged

~~P5~~ There is no concept of the evolution of a state in QFT. This postulate is totally different from before.

The remaining concept for P5

The physical g.p. P is $R^{1,3} \times S_0^+(1,3)$. ($R^{1,3}$ is the g.p. of spacetime translations (for $x \in R^{1,3}$, $x(y) = y + x$).

i.e., $P = \{ (a, A) : a \in R^{1,3}, A \in S_0^+(1,3) \}$ with $(a, A)(b, B) = (a + b, AB)$

This is the g.p. of isometries of the Minkowski spacetime.

The action of $(a, A) \in P$ on $x \in R^{1,3}$ is $(a, A)(x) = a + Ax$

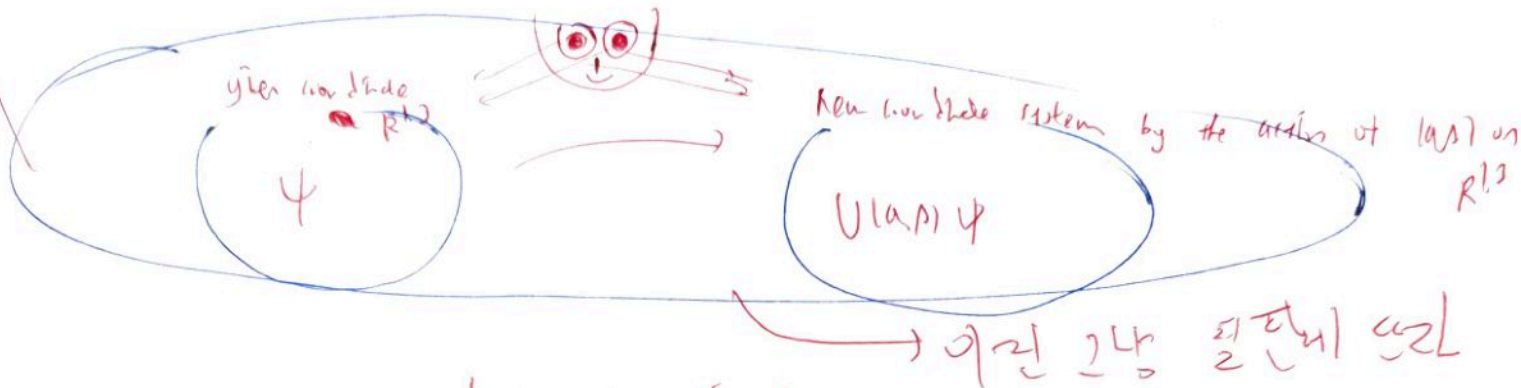
The action of physical g.p. in H is a map $U: P \rightarrow \mathcal{U}(H)$ the space of unitary operators on H .

s.t. $U((a, A)(b, B)) = U(a, A)U(b, B)$ $U(a, A), (1, B)$

Strongly continuous We call the representation 'strongly continuous' if for any $x \in H$, $(a, A) \mapsto U(a, A)x$ is continuous.

So, the revised PS replaces the action of a time evolution g.p. of unitary operators with a strongly continuous unitary representation of the ~~path~~ group $\mathcal{P}$ on $\mathcal{H}$.

P5 Any quantum system with H (now) equipped with a strongly continuous unitary representation U of $\mathcal{P}$ in $\mathcal{H}$.



Lecture 12 (The massive scalar field)

12-1 creation and annihilation on the mass shell

$H = L^2(X_m, d\lambda_m)$, $\mathcal{B} =$ Bosonic Fock space for this H .

Recall $A, A^\dagger$ on $\mathcal{B}$ and $A(t) = \int d\lambda_m t^*(p) a(p)$
 $A^\dagger(t) = \int d\lambda_m t(p) a^\dagger(p)$

$(a(p)\phi)(p_1, \dots, p_n) = \sqrt{n} \phi(p_1, \dots, p_n, p)$, $(a^\dagger(p)\phi)(p_1, \dots, p_{n+1}) = \frac{1}{\sqrt{n+1}} \sum_{i=1}^{n+1} \delta_{p_i}(p) \phi(p_1, \dots, \hat{p}_i, \dots, p_{n+1})$

~~Recall~~ X: We cannot use $\delta(p-p')$ since X_m is not a vector space and so, $p-p'$ may not belong to X_m .

We define two related operator-valued distributions on $\mathcal{B}$.

$$a(\lambda, p) = \frac{1}{\sqrt{2\omega_p}} a(p)$$

$$a^\dagger(\lambda, p) = \frac{1}{\sqrt{2\omega_p}} a^\dagger(p)$$

Commutation relations for $a(\lambda, p)$ and $a^\dagger(\lambda, p)$

$$(1) [a(\lambda, p), a(\lambda', p')] = 0$$

$$(2) [a^\dagger(\lambda, p), a^\dagger(\lambda', p')] = 0 \quad (3) [a(\lambda, p), a^\dagger(\lambda', p')] = (2\pi)^3 \delta^3(p-p')$$

To prove (2),

$$\iint \frac{d^3p}{(2\pi)^3} \frac{d^3p'}{(2\pi)^3} f(p)^* g(p') [a(\lambda, p), a^\dagger(\lambda', p')]$$

$$= \iint \frac{d^3p}{(2\pi)^3} \frac{d^3p'}{(2\pi)^3} \frac{1}{\sqrt{4\omega_p \omega_{p'}}} f(p)^* g(p') [a(p), a^\dagger(p')]$$

$$= \iint d\lambda_m(p) d\lambda_m(p') \sqrt{4\omega_p \omega_{p'}} f(p)^* g(p') [a(p), a^\dagger(p')]$$

$$= \int d\lambda_m(p) 2\omega_p f(p)^* g(p) - 1 = \left(\int \frac{d^3p}{(2\pi)^3} f(p)^* g(p) \right) \cdot 1$$

The identity operator.

Now we introduce the massive scalar free field.

• Q (massive scalar field) is an operator-valued distribution on $\mathcal{S}(\mathbb{R}^3)$ (Schwartz test on $\mathbb{R}^3$).

• The action of Q on a $f \in \mathcal{S}(\mathbb{R}^3)$ is ~~well~~ defined $Q(f) = A(\hat{f}|_{x_m}) + A^+(\hat{f}|_{x_m})$.

where $\hat{f}(p) = \int d^3x e^{i(x,p)} f(x)$.

Now, we move on to the Hamiltonian on Fock space.

• $H :=$ the Hamiltonian for free evolution on $\mathcal{H} = L^2(\mathbb{R}^3, d^3p)$

Recall on $L^2(\mathbb{R})$, $H = \int_{-\infty}^{\infty} dp \cdot \left(\frac{p^2}{2m}\right) a^+(p) a(p)$ (classical theory)

From the Hamiltonian for the evolution in the relativistic setting, $H_0 \psi(p) = p^0 \psi(p)$.

~~the above~~ the above H_0 can be expressed by

$$\begin{aligned} H_0 &= \int_{\mathbb{R}^3} d^3p(p) p^0 a^+(p) a(p) \\ &= \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \sqrt{2\omega_p} a^+(p) \sqrt{2\omega_p} a(p) \\ &= \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \omega_p a^+(p) a(p) \end{aligned}$$

12.4 Particle representation of the free field

• Q (massive scalar free field of mass m), recall $f \in \mathcal{S}(\mathbb{R}^3)$, $\hat{f}(p) = \int d^3x e^{i(x,p)} f(x)$.

$$\Rightarrow Q(f) = A(\hat{f}|_{x_m}) + A^+(\hat{f}|_{x_m})$$

$$= \int d^3p(p) (\hat{f}^*(p) a(p) + \hat{f}(p) a^+(p))$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} (\hat{f}^*(p) a(p) + \hat{f}(p) a^+(p))$$

$$= \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (\hat{f}^*(p) a(p) + \hat{f}(p) a^+(p))$$

plug $\Rightarrow$ $Q(f) = \int_{\mathbb{R}^3} d^3x f(x) \left[\int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (e^{-i(x,p)} a(p) + e^{i(x,p)} a^+(p)) \right]$

$Q(x)$. (particle representation of Q).

12.5 Normal ordering

For example, $a^+(p) a^+(q) = a^+(q) a^+(p)$ (50 page 241-242)

Normal ordering will play an important role in the construction of interaction Hamiltonian later.

12.6 Expressing the Hamiltonian using the free fields

• $x = (t, \mathbf{x})$ and $\mathbf{x} = (x^1, x^2, x^3)$.

Recall $\phi(x) = \int_{R^3} \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{\sqrt{2\omega_p}} (e^{-i(x,p)} a(p) + e^{i(x,p)} a^\dagger(p))$.

$\Rightarrow H_0 = \int_{R^3} \frac{d^3p}{(2\pi)^3} \omega_p a^\dagger(p) \cdot a(p) = \frac{1}{2} \int_{R^3} d^3x : (\partial_t \phi(x))^2 + \sum_{i=1}^3 (\partial_i \phi(x))^2 + m^2 \phi(x)^2 :$

15) RHS = $\frac{1}{2} \int \frac{d^3x d^3p d^3q}{(2\pi)^6} = \left[-\frac{\sqrt{\omega_p \omega_q}}{2} (e^{i(x,p)} a(p) - e^{-i(x,p)} a^\dagger(p)) (e^{i(x,q)} a(q) - e^{-i(x,q)} a^\dagger(q)) \right.$
 $+ \frac{1}{2\sqrt{\omega_p \omega_q}} (i p \cdot e^{i(x,p)} a(p) - i p \cdot e^{-i(x,p)} a^\dagger(p)) \cdot (i q \cdot e^{i(x,q)} a(q) - i q \cdot e^{-i(x,q)} a^\dagger(q))$
 $\left. + \frac{m^2}{2\sqrt{\omega_p \omega_q}} (e^{i(x,p)} a(p) + e^{-i(x,p)} a^\dagger(p)) (e^{i(x,q)} a(q) + e^{-i(x,q)} a^\dagger(q)) \right] =$

$= \frac{1}{4} \int \frac{d^3p}{(2\pi)^3} \cdot \frac{1}{\omega_p} : \left[(-\omega_p^2 + p^2 + m^2) \cdot (a(p) a^\dagger(p) + a^\dagger(p) \cdot a(p)) + \right.$
 $\left. (\omega_p^2 + m^2 + p^2) \cdot (a(p) \cdot a^\dagger(p) + a^\dagger(p) \cdot a(p)) \right] :$

$= \frac{1}{4} \int \frac{d^3p}{(2\pi)^3} \cdot \omega_p : (a^\dagger(p) \cdot a(p) + a(p) a^\dagger(p)) :$

$= \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \cdot \omega_p \cdot (a^\dagger(p) a(p))$

□

-X.

Here, from the second to the third, we use the relation

$\left(\begin{aligned} \int d^3x \cdot e^{i(x,p)} &= (2\pi)^3 \cdot \delta^{(3)}(p) \\ \int d^3x \cdot e^{i(x,q)} &= (2\pi)^3 \cdot \delta^{(3)}(q) \end{aligned} \right)$ to integrate it w.r.t d^3x .